YASHWANTH GOWDA S N

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EDUCATION

KS Institute of Technology (KSIT), Bengaluru **Expected May 2029**
B.E. Computer & Communication Engineering CGPA: **8.52/10.0**

SKILLS

Languages: Java, Python, C, C++, SQL, JavaScript
Backend: Spring Boot, Spring Cloud Gateway, Spring Security, FastAPI, REST APIs, WebSockets, Kafka
Frontend & Databases: React.js, HTML, CSS, MySQL, MongoDB, Redis, ChromaDB
AI/ML: TensorFlow, TFLite Micro, INT8 Quantization, LangChain, RAG, Gemini API, OpenAI API, Ollama
Embedded/IoT: ESP32, NXP FRDM-MCXN236, ESP-NOW, UART, SPI, Arduino
DevOps & Tools: Docker, Kubernetes, Git, Linux, Maven, Gradle, Postman
Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems

PROJECTS

RESq Mesh – Edge AI & Autonomous Hardware Telemetry Mesh GitHub
C/C++, Python, TFLite Micro, INT8 Quantization, NXP MCXN236, ESP32, ESP-NOW, HMAC-SHA256

- Built an edge network-health prediction system on NXP FRDM-MCXN236 (Cortex-M33) using an INT8-quantized TFLite Micro model: **10.3 ms** inference, **13,572-byte** tensor arena.
- Implemented ESP32 peer-to-peer forwarding over ESP-NOW with offline state sync using causal sequence tracking and HMAC-SHA256 integrity checks.

Cloud Native API Gateway
Java, Spring Boot, Spring Cloud Gateway, Redis, Kafka, Docker, Kubernetes, JWT, Resilience4j

- Developed an API gateway with Spring Cloud Gateway, JWT authentication, and Redis-based rate limiting.
- Integrated Kafka for async communication and Resilience4j for fault tolerance; Dockerized for Kubernetes-ready deployment.

Persistent Memory AI Assistant
Python, FastAPI, LangChain, ChromaDB, RAG, Gemini API, OpenAI API, Docker

- Built an AI assistant with RAG-based semantic search and contextual memory retrieval for multi-session conversations.

EcoSync AI – 3rd Place, EcoSpark 2026
ESP32, Arduino, C++, React.js, Spring Boot, MySQL, Gemini API

- Built an AI-powered IoT composting platform with ESP32 sensors, Spring Boot, and React for real-time monitoring and compost analysis.

ACHIEVEMENTS

- **1st Place among 102 teams** at **HACKFEST 2026**, 24-hour National-Level Hackathon, Sai Vidya Institute of Technology; won INR 20,000.
- **1st Place – Overall Winner**, **MLH Hacktoberfest Hack Day Bengaluru 2026**; won INR 10,000 and an Arduino UNO Q board.
- **1st Place**, ICARUS – CanSat Telemetry Analysis Competition, RV University.
- **2nd Place**, GitSetGo! open-source Git/GitHub relay race, RV University.
- Received the **Suzlon Young Brilliance Award 2026**.
- Solved **270+ LeetCode problems**; contest rating **1587**.